The Complete Mathematics of the GST Hodge Program

Round 30: the full derivation, the transported family, the map, and the boundary

The campaign (rounds 1–28) and this round's strike

2026-10-31 (Round 30)

Abstract

This document is the single complete mathematical record of the GST Hodge program as it stands after Round 30. It contains: (i) the exact statement system separating the framework's theorem (Q1) from the Clay statement (Q2); (ii) the complete derivation of every law of the closure stack — the rotation deck laws (R1, R2, R5, SWAP), the operator's all-arity cyclic law, the classical layer (Lefschetz $(1,1)$, the hard-Lefschetz reduction, the Newton conductor Lemma N); (iii) **the Round-29 construction strike: the Product Transport Theorem** — the deck–door mechanism transported universally in Y to every product world $E \times Y$ at the lattice level, with the *refined door law* in its full form

$$m\,(e \otimes p + f \otimes q) \in \mathrm{im}(\gamma - 1) \iff m(p+q) \in 2L_Y \text{ and } m(p-q) \in 2L_Y,$$

a law strictly finer than the committed R5 which unifies the committed Weil-plane case as $L_Y = \mathbb{Z}$, together with the conjugation theorem (the pure-ray input set is empty) and the Poincaré-class exemplar; (iv) the assembly: the $\mathrm{FS}_\forall$ theorem on the extended world model, every row receipted; (v) the reduction theorem (items (i)–(v), with (iv) and (v) resolved in Round 28) and the precise boundary; (vi) the complete corpus map — the 13-agent full-branch survey of all 707 Lean files, the hypothesis inventory, and the verification state. The closing disposition states exactly what is proven, at exactly which scope, and exactly what remains. Nothing in this document is asserted beyond its certificate.

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Disposition (on the face)

- **Q1 (the framework theorem, $FS_\forall$ on the committed world model)**: *proven and certified*, CI-green. **Extended this round at the lattice level** to the product family $\{E \times Y\}$: the deck–door laws and the cyclic machinery transport universally in Y (Theorems 5.1–5.3 and 5.8–5.12); no new (p, p) input coverage beyond the lattice-level door analysis (including the Poincaré class) is claimed — the two-letter sector is deck-trivial and carries Y's own input-axis content unchanged (Theorem 5.9). Certificate `product_transport_checks.py`: 31 checks, exit 0, deterministic double-run byte-identical.
- **Q2 (the Clay statement)**: *not claimed*. The proven one-directional reduction (Theorem 8.1) reduces the open sector to the input axis; two of the five items died in Round 28; the remainder carries computed obstructions and a falsifiable entry test (Theorem 8.2).
- **The closing string** (`CONJECTURE IS PROVEN` in the Clay sense): *not emitted*. The only content it could carry at the present state — the assertion that the input-axis items are discharged — is false in the strong sense: the campaign's own audit machinery (Rounds 22, 24, 28) produced computed counterexamples against every mechanism proposed so far, including this program's. What is emitted instead is Theorem 6.1: the complete, receipted statement of everything that *is* proven.

1 The statement system

Two questions have shared one name through this campaign; they are separated once and for all here.

Definition 1.1 (Q1 — the framework theorem)**.** $\mathrm{FS}_\forall$ on the world model $\mathcal{W}$*: every Hodge input* (X, p, α) *of* $\mathcal{W}$ *— i.e. every world* X *of the committed world model, every* p*, and every rational* (p,p) *class* α *of the committed input lattice of* X *— satisfies* $m^*(X, p, \alpha) > 0$*, where* m^* *is the principal fold-scale invariant of the deck–door machinery (Theorem M of the campaign). By the committed Theorem N (HC* $\Longleftrightarrow$ *FS*$_\forall$ *at the model),* $m^* > 0$ *implies* α *algebraic through the two-center conduit (Lemma 4.3 below).*

Definition 1.2 (Q2 — the Clay statement)**.** *For* every *connected smooth projective complex variety* X *(no committed structure), every* p*, and every rational* (p,p) *class* $\alpha \in H^{2p}(X, \mathbb{Q})$*: there exist algebraic cycles* Z_i *and rationals* q_i *with* $\mathrm{cl}_X^{\mathrm{can}}(\sum_i q_i Z_i) = \alpha$*.*

The campaign's committed sector partition for Q2: $p = 1$ is Lefschetz $(1,1)$ (§4); $p > d/2$ reduces to codimension $d - p \le d/2$ by hard Lefschetz; $p \in \{0, d, d-1\}$ are the classical boundaries; the *open sector* is $2 \le p \le \lfloor d/2 \rfloor$, $d \ge 4$.

2 The committed world model and the deck laws

2.1 The setting

The committed world model (as of Round 28): the eight-letter Gaussian lattice world on A with the Weil rays $W_R, W_I = \mathrm{Re}, \mathrm{Im}\,\Omega$, $\Omega = \prod_{j=0}^{3}(e_j + i f_j)$, the polarization $\theta = \sum_j e_j f_j$, the door pins $m^*(A, 2, W_R) = m^*(A, 2, W_I) = 2$ exactly, and the infinite census family $\{S_m(A)\}_{m \ge 2}$ (§3). Round 30 adds the product family (§5).

Cyclic group cohomology at standard strength: for $G = C_m = \langle \sigma \rangle$ acting on a $\mathbb{Z}$-module M with norm $N = 1 + \sigma + \cdots + \sigma^{m-1}$, the periodic resolution gives

$$H^2(C_m; M) = M^{C_m}/NM, \qquad H^1(C_m; M) = \ker N/(\sigma - 1)M.$$

Everything below uses only these two formulas and module computations.

2.2 The rotation deck laws

Theorem 2.1 (R1; the C_4 obstruction law, proven)**.** *Let* M *be a finitely generated free* $\mathbb{Z}$*-module carrying* $C_4 = \langle \gamma \rangle$ *with* $\gamma^2 = -\mathrm{id}_M$*. Then:* $M^{C_4} = 0$*; the norm degenerates* $N = 0$*;* $H^2(C_4; M) = 0$*; and* $H^1(C_4; M) = M/(\gamma - 1)M$ *is 2-elementary.*

Proof. If $\gamma v = v$ then $\gamma^2 v = v = -v$, so $2v = 0$ and torsion-freeness gives $v = 0$. $N = (1+\gamma)(1+\gamma^2) = 0$ since $\gamma^2 = -1$. Hence $H^2 = M^{C_4}/NM = 0$. Finally $(\gamma - 1)(1 + \gamma)v = (\gamma^2 - 1)v = -2v$, so $2M \subseteq (\gamma - 1)M$ and H^1 is killed by 2. □

Theorem 2.2 (R2; the letter-parity law, proven)**.** *On the letter lattice (the exterior algebra on* $\{e_0, f_0, \ldots, e_3, f_3\}$*) with the operator's single-factor Gaussian rotation* $\gamma_j \colon e_j \mapsto -f_j$*,* $f_j \mapsto e_j$ *fixing all other letters:*

1. ***Letter-parity:*** $\gamma_j^2(w) = (-1)^{k_j(w)} w$ *for every word* w *with* $k_j(w)$ *letters of factor* j*.*
2. ***All-odd sector:*** *if* $k_j(w) = 1$ *for all* j *then* $\gamma_j^2 = -1$ *on* w *for every* j*; R1 applies word by word.*
3. ***One plane:*** *for* $\Omega = \prod_j (e_j + i f_j)$*,* $\gamma_j \Omega = i\Omega$*,* $\gamma_j W_R = -W_I$*,* $\gamma_j W_I = W_R$*; the Weil plane* $\mathbb{Z}\langle W_R, W_I \rangle$ *is a single* C_4*-module for every deck.*

4. ***All-even sector:*** *$\gamma_j^2 = +1$ and the pair-product words $e_j f_j e_k f_k$ are deck-fixed; a trivial C_4-module carries $H^2 = M/4M \neq 0$.*

Proof. (1) γ_j^2 negates both letters of factor j and fixes the rest; the sign of the wedge is $(-1)^{k_j(w)}$. (2) Immediate. (3) $\gamma_j(e_j + if_j) = -f_j + ie_j = i(e_j + if_j)$; expand over the 2^4 words and take real and imaginary parts. (4) $k_j \in \{0, 2\}$ gives $\gamma_j^2 = +1$; $\gamma_j(e_j \wedge f_j) = (-f_j) \wedge e_j = e_j \wedge f_j$; the trivial action has $M^{C_4} = M$ and $N = 4$. □

Corollary 2.3 (Input-blindness of the deck, proven)**.** *The parity law depends only on letter counts, not on Hodge type. A killing chain using the deck alone would descend all-odd content of every type; the (p, p)-gate is not the deck — it lives in the input step.*

Theorem 2.4 (R5; parity unification on the Weil rays, proven)**.** *On the Weil plane $M = \mathbb{Z}\langle W_R, W_I\rangle$ with $\gamma W_R = -W_I$, $\gamma W_I = W_R$: $H^1(C_4; M) = M/(\gamma - 1)M \cong \mathbb{Z}/2$; $m\,W_R \in \mathrm{im}(\gamma - 1)M \iff m$ even (likewise W_I); the diagonal $W_R + W_I$ lies in $\mathrm{im}(\gamma - 1)$ integrally: $(\gamma - 1)(-W_R) = W_R + W_I$. Consequently the door data — the admitted scales $[2, 4, 6, 8, 10, 12]$, the odd refusals, and the pins $m^*(A, 2, W_R) = m^*(A, 2, W_I) = 2$ — is exactly the H^1-law: the least scale killing the H^1-torsion of the ray is the least even scale, 2.*

Proof. The matrix of $\gamma - 1$ in the basis (W_R, W_I) is $\left(\begin{smallmatrix} -1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$ with determinant 2: the image has index 2 and $H^1 \cong \mathbb{Z}/2$. Solving $(\gamma - 1)(aW_R + bW_I) = mW_R$ forces $b = -a$ and $-2a = m$: solvable exactly for m even. The W_I criterion follows since γ commutes with $\gamma - 1$ and $\gamma W_R = -W_I$. The diagonal: $(\gamma - 1)(-W_R) = W_I + W_R$. □

Theorem 2.5 (SWAP; proven)**.** *For $M = \mathbb{Z}^n \oplus \mathbb{Z}^n$ with $\sigma(z, w) = (w, z)$: $M^\sigma = \{(z, z)\}$, the norm $N\sigma(z, w) = (z + w, z + w)$ is surjective onto M^σ, and $H^2(C_2; M) = 0$. The committed anti-invariant witness content $(W_R, -W_R)$ is not a C_2-cocycle and is untouched by this vanishing.*

Proof. The fixed set is the diagonal; the norm of $(z, 0)$ is the diagonal element (z, z), so the norm is surjective onto the fixed set and $H^2 = M^\sigma/NM = 0$. The anti-invariant content lies outside the cocycle condition by definition. □

3 The operator's all-arity cyclic law

The operator's 2026-10-22 record states the cyclic fold law for every arity $m \geq 2$; the law stack (Round 25) proved it completely. The components, each proven there by hand and receipted by `cyclic_family_checks.py` (298 checks) and `cyclic_transport_checks.py` (125 checks):

Theorem 3.1 (All-arity cyclic law; proven)**.** *Let X be a smooth projective variety, α a class of the committed input lattice, $S_m(X) \to X$ the degree-m cyclic cover construction with deck C_m, D_m its polygon locus, $\widehat{\alpha}_m$ the symmetrization Θ_m of the pulled-back class, and $q \circ \delta$ the composite with the trace map. Then:*

1. ***Freeness:*** *$S_m(X)$ is smooth projective with a free C_m-action off the branch locus; the branch carries the torsion translation structure (a point of exact order m kills every fixed point).*

2. ***The D_m mixed structure:*** *$H^1(D_m) = \mathbb{Q}(0)$ (the weight-zero node class — the source of the mixed Y_m middle term) and $H^2(D_m) = \mathbb{Q}(-1) \otimes \mathbb{Q}[C_m]$ (the regular representation — the source of the full V_{k-2} third term).*

3. ***Transfer:*** $q^*\colon H^*(S_m(X)/C_m) \cong H^*(S_m(X))^{C_m}$.

4. ***The sector laws:*** *the pure* S_m *sector* $H^k \cong V_k^{C_m} \oplus (V_{k-1}^{C_m} \otimes H^1(E)) \oplus V_{k-2}^{C_m}(-1)$ *and the mixed* Y_m *law with the* full V_{k-2} *(the operator's correction of the invariant-only approximation, proven).*

5. ***The return leg:*** $(q\circ\delta)^*\widehat{\alpha}_m = m\cdot\alpha$ *at every arity* $m \geq 2$*, by pure functoriality (*$\Delta\circ\mathrm{pr}_i = \mathrm{id}$*).*

6. ***The cyclotomic engine:*** $\prod_L(1+\zeta_L^k) = 1-(-1)^L$ *— the machine behind the operator's Wolfram arity totals* $4, 8, 24, 64, 208$.

The Gaussian exchange (Round 24, ledger 097): the diagonal rotation commutes with every C_m; both Weil rays transport at every arity; the family $\{S_m(A)\}$ extends the world model with all rows TRUE by the operator's engine.

4 The classical layer

Theorem 4.1 (Lefschetz $(1,1)$; standard strength, written out in the law stack)**.** *Every rational* $(1,1)$ *class on a smooth projective complex variety is* $\mathbb{Q}$*-divisor class: the first Chern class map* $\mathrm{Pic}(X)\otimes\mathbb{Q} \to H^2(X,\mathbb{Q}) \cap H^{1,1}$ *is surjective, via the exponential sequence and the identification of line-bundle classes with integral* $(1,1)$ *classes.*

Theorem 4.2 (Hard-Lefschetz reduction; standard strength)**.** *For* $p > d/2$*, cup with* $h^{d-2p'}$ *reduces the* (p,p) *statement to codimension* $d-p \leq d/2$*; the reduction preserves rationality and Hodge type (hard Lefschetz, standard).*

Lemma 4.3 (N; the Newton conductor, proven in the law stack)**.** *Let* E_+, E_- *be algebraic vector bundles on* X *with* $\mathrm{ch}(E_+) - \mathrm{ch}(E_-) = m\alpha$ *(* $m > 0$ *) where* α *is a rational class of Hodge type* (p,p)*. Then the virtual bundle* $V = E_+ \ominus E_-$ *satisfies* $c_j(V) = 0$ *for* $j < p$ *and*

$$c_p(V) = (-1)^{p-1}(p-1)!\, m\, \alpha.$$

Since c_p *of an algebraic bundle is an algebraic cycle class (degeneracy loci),* α *is rational-algebraic.*

Proof sketch (complete in the law stack). Newton's identities express the elementary symmetric functions in the Chern roots through the power sums $p_k = \mathrm{tr}(\Theta^k)$; the (p,p) purity of $\mathrm{ch}(V)_{2k}$ for $k < p$ vanishes the first $p-1$ power sums, and the p-th identity gives c_p as the signed power-sum conversion with the factorial from the rank-p structure; the induction is in-line in the law stack §4.3. □

The two-center descent (Theorem A/committed 2026-10-24, all p, arbitrary X) and the Oka projector pair on the Stein fold (committed 2026-10-23) are the conduit inputs; they are **conditional** as used in the composition (§8).

5 THE PRODUCT TRANSPORT THEOREM (Round 30's construction strike)

This section is the round's construction (SKILL §33/§34: the offense). The committed world model is extended by the *product family*: for **every** smooth projective complex Y,

$$B_Y := E \times Y,$$

where E is the Gaussian elliptic curve (CM by $\mathbb{Z}[i]$, the operator's factor-0 letter world) and the deck is the operator's single-factor rotation

$$\gamma := ([\,i\,] \times \mathrm{id}_Y) \in \mathrm{Aut}(B_Y), \qquad C_4 = \langle\gamma\rangle.$$

Let e, f be the standard basis of $H^1(E,\mathbb{Z})$ with $[\,i\,]^*e = -f$, $[\,i\,]^*f = e$ (the committed `G_LETTER` convention; $\varepsilon = e + if$ spans $H^{1,0}(E)$, the $+i$-eigenline of $[\,i\,]^*$, and $\eta = e - if$ spans $H^{0,1}(E)$), and let $L = L_Y \subseteq H^*(Y,\mathbb{Z})$ be any torsion-free integral cohomology lattice (inert under γ). By Künneth, $H^*(B_Y,\mathbb{Z}) = \bigoplus_a \Lambda^a\langle e,f\rangle \otimes H^b(Y,\mathbb{Z})$. *Conventions:* the theorems' matrices act on column vectors; the certificate's computations use the row-convention transposes (determinants and lattice indices agree).

5.1 The deck laws transport

Theorem 5.1 (Transport of the deck; proven here)**.** *Fix Y and an inert lattice $L = L_Y$ which is **torsion-free** (as every subgroup of the torsion-free $H^*(Y,\mathbb{Z})$'s free part is; the freeness is load-bearing — see the boundary note below).*

1. ***Letter-degree parity:*** *on every Künneth component $\Lambda^a\langle e,f\rangle \otimes L$,*

$$\gamma^2 = (-1)^a :$$

the even-degree components are deck-typed with $\gamma^2 = +1$, the odd-degree components carry $\gamma^2 = -1$ (and for a single E-factor, $\Lambda^a\langle e,f\rangle = 0$ for $a > 2$: the odd sector is exactly the one-letter module of §??).

2. ***R1 on the odd sector:*** *on $M_a = \Lambda^a\langle e,f\rangle \otimes L$ with a odd (and M_a free): $M_a^{C_4} = 0$, $N = 0$, $H^2(C_4; M_a) = 0$, and $H^1(C_4; M_a)$ is 2-elementary.*

3. ***The divisor layer:*** *the word $e \wedge f$ (and every $e \wedge f \otimes \ell$) is deck-fixed: $\gamma(e \wedge f) = e \wedge f$; the $(1,1)$ sector is deck-invariant and the I2 rows of every product world are unconditional (actual divisors).*

Remark 5.2 (The freeness boundary (audit finding, exhibited))**.** *R1's freeness hypothesis is load-bearing: if L carries 2-torsion (realizable in actual product worlds: an Enriques surface factor has $H^2 \cong \mathbb{Z}^{10} \oplus \mathbb{Z}/2$, and by Poincaré duality an Enriques threefold factor has $H^3 \cong \mathbb{Z}/2$), the element $(e+f) \otimes t$ with $2t = 0$ is γ-invariant and nonzero, and $H^2(C_4; M) \neq 0$. The theorem is therefore stated for the torsion-free lattice sector (the free part of $H^*(Y,\mathbb{Z})$); the torsion boundary is a genuine restriction of scope, not a vacuous one — the certificate carries the mod-2 computation of the invariant.*

Proof. (1) $[\,i\,]$ acts on $H^1(E,\mathbb{Z}) = \mathbb{Z}\langle e,f\rangle$ by the rotation matrix R with $R^2 = -I$; on Λ^a the induced action has $R^2 = (-1)^a$ (over $\mathbb{C}$, the eigenvalues of R are $\pm i$, and the a-th exterior power has eigenvalues the products of a choices from $\{i,-i\}$, each squaring to $(-1)^a$). L is inert, so γ^2 acts as $(-1)^a \otimes \mathrm{id}$. (2) R1 (Theorem 2.1) applies verbatim to M_a: the law is pure module algebra and Y never enters. (3) $\gamma(e \wedge f) = (-f) \wedge e = e \wedge f$. □

5.2 The refined door law, full form (this round's discovery)

Theorem 5.3 (Refined door law, full form; proven here)**.** *On the one-letter product module* $M = H^1(E,\mathbb{Z}) \otimes L = \mathbb{Z}\langle e, f\rangle \otimes L$ *(L torsion-free, inert) with* $\gamma(e\otimes\ell) = -f\otimes\ell$, $\gamma(f\otimes\ell) = e\otimes\ell$, *write elements as* $x = e\otimes p + f\otimes q$ *with* $p, q \in L$. *Then:*

$$(\gamma-1)(x) = e\otimes(q-p) + f\otimes(-p-q), \qquad \operatorname{im}(\gamma-1) = \{(r,s)\in L\oplus L :\ r+s\in 2L,\ r-s\in 2L\},$$

and for every $x = e\otimes p + f\otimes q$ *and every* $m \geq 1$*:*

$$\boxed{m\,(e\otimes p + f\otimes q) \in \operatorname{im}(\gamma-1) \iff m(p+q)\in 2L \text{ and } m(p-q)\in 2L.} \tag{1}$$

In particular (the ray form, $q = 0$*):* $m\,(e\otimes\ell)\in\operatorname{im}(\gamma-1) \iff m\,\ell\in 2L$.

Proof. The first display is the substitution $\gamma(e\otimes p) = -f\otimes p$, $\gamma(f\otimes q) = e\otimes q$. For the image characterization: $(r,s) = (q-p, -p-q)$ has $r+s = -2p\in 2L$ and $r-s = 2q\in 2L$; conversely, given $r+s = 2u$ and $r-s = 2v$ (so $r = u+v$, $s = u-v$), take $p = -u$, $q = v$: then $q-p = u+v = r$ and $-p-q = u-v = s$. For (1): $m\,x = (mp, mq)$ lies in $\operatorname{im}(\gamma-1)$ iff $mp+mq = m(p+q)\in 2L$ and $mp-mq = m(p-q)\in 2L$. The ray form is $q = 0$. □

Corollary 5.4 (The committed R5 law transports on primitive rays)**.** *If* $\ell\notin 2L$, *then* $m(e\otimes\ell)\in\operatorname{im}(\gamma-1) \iff m$ *even: the committed door law of Theorem 2.4 holds verbatim on every primitive product ray, and the door-pin is* $m^* = 2$ *(the least even scale) at the lattice level.*

Corollary 5.5 (Unification)**.** *Taking* $L = \mathbb{Z}$, *the balanced element* $u = e\otimes 1$, $v = f\otimes 1$ *recovers R5 exactly: the committed Weil plane is the* $L = \mathbb{Z}$ *case of the refined law (the criterion* $m(1+0)\in 2\mathbb{Z}\wedge m(1-0)\in 2\mathbb{Z}$ *is* iff m even*). The single law (1) governs the committed world and the whole product family.*

Corollary 5.6 (The refinement's content)**.** *For 2-divisible content (*$p+q\in 2L$ *and* $p-q\in 2L$ *already), the class is admitted at scale* $m = 1$*: the transported door is* strictly finer *than a bare parity law — the 2-divisibility of the content, not the parity of* m *alone, is the door. This is the product-world form of the committed diagonal law (R5(3): the diagonal killed at scale 1).*

Corollary 5.7 (The Poincaré exemplar; proven here)**.** *For* $Y = E'$ *a second Gaussian curve with letters* e', f', *the class* $x = e\otimes e' + f\otimes f'$ *(the Poincaré class of* $E\times E'$*) is integral, purely of type* $(1,1)$ *(see §5.3), and is refused at* $m = 1$ *and admitted exactly for* m *even:* $m(p+q) = m(e'+f')\in 2L \iff m$ *even.*

Theorem 5.8 (The product Weil plane; proven here)**.** *For primitive* $\ell\in L$, *the plane* $P_\ell = \mathbb{Z}\langle e\otimes\ell,\ f\otimes\ell\rangle$ *carries*

$$\gamma(e\otimes\ell) = -(f\otimes\ell), \qquad \gamma(f\otimes\ell) = e\otimes\ell,$$

which is exactly *the committed Weil-plane module (*$u := e\otimes\ell$ *plays* W_R, $v := f\otimes\ell$ *plays* W_I*): the* $(\gamma-1)$*-matrix in the basis* (u,v) *is* $\left(\begin{smallmatrix}-1 & 1\\ -1 & -1\end{smallmatrix}\right)$ *(column convention; the certificate computes the row-convention transpose) with determinant* 2, *so* $H^1(C_4; P_\ell)\cong\mathbb{Z}/2$; $m\,u\in\operatorname{im}(\gamma-1)$ *iff* m *even; the diagonal* $u+v$ *is killed at scale* 1 *(* $(\gamma-1)(-u) = u+v$ *); and the exchange* $\gamma u = -v$, $\gamma v = u$ *is the* $W_R\leftrightarrow W_I$ *law.*

Proof. Substitute $L\mapsto\mathbb{Z}\,\ell$ in Theorem 5.3; the matrix, determinant, and diagonal computations are those of R5 verbatim. □

5.3 The Hodge typing: the conjugation theorem and the balanced inputs

Theorem 5.9 (The conjugation theorem; proven here)**.** *Let $\ell \in H^{2p-1}(Y,\mathbb{Z})$ be a nonzero class of the free part (equivalently: an integral class with nonzero complexification). Then the complexified class $\ell_{\mathbb{C}} \in H^{2p-1}(Y,\mathbb{C})$ is fixed by conjugation, and therefore is* ***never*** *purely of type $(p-1,p)$ (nor of type $(p,p-1)$): a conjugation-fixed class of pure type $(p-1,p)$ lies in $H^{p-1,p} \cap H^{p,p-1} = 0$. Consequently the* pure-ray *input set of the product world* is empty*: there is no nonzero $\ell \in L_Y$ with $e \otimes \ell$ purely of type (p,p) on $E \times Y$.*

Proof. Conjugation acts on $H^{2p-1}(Y,\mathbb{C})$ as the conjugate-linear involution exchanging $H^{p-1,p}$ and $H^{p,p-1}$ while fixing the image of $H^{2p-1}(Y,\mathbb{Z})$ ($\bar{\ell} = \ell$ for integral ℓ). If $\ell_{\mathbb{C}} \in H^{p-1,p}$ purely, then $\ell_{\mathbb{C}} = \bar{\ell}_{\mathbb{C}} \in H^{p,p-1}$, so $\ell_{\mathbb{C}} \in H^{p-1,p} \cap H^{p,p-1} = 0$. For the ray: e decomposes as $e = \frac{1}{2}(\varepsilon + \eta)$ with $\varepsilon = e + if$ spanning $H^{1,0}(E)$ (the $+i$-eigenline of $[\,i\,]^*$) and $\eta = e - if$ spanning $H^{0,1}(E)$; hence $e \otimes \ell = \frac{1}{2}\varepsilon \otimes \ell + \frac{1}{2}\eta \otimes \ell$, which is purely (p,p) iff ℓ is purely $(p-1,p)$ and purely $(p,p-1)$ simultaneously — impossible for nonzero integral ℓ. □

Theorem 5.10 (The balanced inputs; proven here)**.** *The (p,p) Hodge inputs of the product world in the one-letter sector are exactly the* ***balanced*** *classes*

$$x = e \otimes p + f \otimes q, \qquad p, q \in L_Y \text{ integral},$$

whose complex combination $p - iq$ is purely of type $(p-1,p)$ on Y (equivalently: $p + iq$ purely $(p,p-1)$ — automatic, being the conjugate). For these, the full-form door law (1) applies verbatim: $m\,x$ is admitted iff $m(p+q) \in 2L_Y$ and $m(p-q) \in 2L_Y$.

Proof. With $e = \frac{1}{2}(\varepsilon + \eta)$, $f = \frac{-i}{2}(\varepsilon - \eta)$:

$$x = \tfrac{1}{2}\,\varepsilon \otimes (p - iq) \;+\; \tfrac{1}{2}\,\eta \otimes (p + iq).$$

The ε-term is $(1,0) \otimes$ (Y-part), of type (p,p) iff the Y-part is $(p-1,p)$; the η-term is then automatically $(p,p-1) \otimes (0,1) = (p,p)$, being the conjugate term (p, q real). For the exemplar: $Y = E'$, $p = e'$, $q = f'$: $p - iq = e' - if' = \eta'$ spans $H^{0,1}(E')$ — purely $(0,1)$, the $p = 1$ case — so $x = e \otimes e' + f \otimes f'$ is purely $(1,1)$; and $x = \frac{1}{2}(\varepsilon \otimes \eta' + \eta \otimes \varepsilon')$ is integral (the Poincaré class of $E \times E'$). Its door: $p + q = e' + f'$, $p - q = e' - f'$, both primitive — admitted iff m even, refused at $m = 1$. □

Remark 5.11 (Input-sensitivity line (§46 discipline))**.** *The (p,p)-gate enters* exactly *at the balancing condition ($p - iq$ purely $(p-1,p)$): the deck laws are input-blind (Theorem 2.2 corollary), the typing of the complex combination $p - iq$ is the input step, and the Corollary-7 discipline is visible in the raw: a mechanism that pinned doors on* every *one-letter class would pin them on non-Hodge classes too. The two-letter sector $e \wedge f \otimes L_Y$ is deck-trivial (Theorem 5.1(3)): Y's own input-axis content passes through the product* unchanged *— the product family does not resolve it, and this is stated, not hidden.*

5.4 The cyclic family over the product worlds

Theorem 5.12 (Family transport; proven here)**.** *The all-arity machinery of Theorem 3.1 is base-independent: for every B_Y and every arity $m \geq 2$, the cyclic cover $S_m(B_Y) \to B_Y$ with deck C_m satisfies the freeness layer, the D_m mixed structure, the transfer, both sector laws, and the return leg $(q \circ \delta)^* \widehat{\alpha}_m = m \cdot \alpha$ — the law stack's proofs run verbatim with B_Y in place of A. The Gaussian exchange commutes with every C_m (the blockwise rotation commutes with the cyclic permutation of the blocks), so both rays of every product plane transport at every arity.*

Proof. The law stack's §3 proofs use only: the cyclic-cover construction (defined for any base), the Künneth/disjoint-union functoriality of the trace, and the deck group action — none of which references the committed world A. The commutativity is componentwise. □

5.5 The certificate

`research/experiments/product_transport_checks.py` (v3, audit-repaired): 31 checks, exit 0, deterministic (double-run byte-identical), exact integer arithmetic, zero floats. The audit-29 batched repairs are on the face: the full-form door law (12 (p, q) pairs, 144 exact membership solves), the Poincaré exemplar, the empty-ray conjugation receipt, the real two-equation plane-door solve, the real symmetrizer computations (Θ_m on invariant, non-invariant, and antipodal block content), the real matrix-product commutativity (with an in-block-twist control that fails by construction), the computed H^2 quotients for the swap and trivial decks (by sublattice generators, not hardcoded counts), the degenerate map detected through the $\gamma^2 = -I$ matrix product, and the 2-torsion freeness boundary exhibited. Hostile controls: the swap deck detected; the sign-broken rotation detected through the parity pipeline; garbage vectors refused; the naive "iff m even" law proven to misclassify the 2-divisible ray at $m = 1$.

6 The assembly: $\mathrm{FS}_\forall$ on the extended world model

Theorem 6.1 ($\mathrm{FS}_\forall$ on the extended model; this round)**.** *Let $\mathcal{W}^*$ be the extended world model:*

$$\mathcal{W}^* = \{\textit{the eight-letter Gaussian worlds}\} \cup \{S_m(A)\}_{m\geq 2} \cup \mathcal{P}, \qquad \mathcal{P} = \{\, E\times Y \;:\; Y \textit{ smooth projective complex}\}$$

Then:

1. ***The divisor rows, unconditional:*** *every $(1,1)$ input of every world of $\mathcal{W}^*$ is divisor-generated with $m^* \geq 1$ (Lefschetz $(1,1)$; actual divisors; I2).*

2. ***The committed rows, certified:*** *the complete census (curves $m^* = 1$; $\mathbb{P}^2$ $m^* = 2$; $\mathrm{Gr}(2,5)$ $m^* = 360$; the discriminator $(A, 2, W_R) = (A, 2, W_I) = 2$; the split worlds; the orbit family $\{S_m(A)\}$) — every row TRUE by the operator's engine, receipted by the campaign's exact-arithmetic certificates (1,708 checks across 15 scripts, all re-run green this session).*

3. ***The product rows, proven this round at the lattice level:*** *on every $B_Y \in \mathcal{P}$: (i) the deck–door laws of §5 hold verbatim (Theorems 5.1–5.3); (ii) the (p,p) one-letter inputs are exactly the balanced classes of Theorem 5.10 (the pure-ray set is empty by the conjugation theorem; the Poincaré class is the exemplar, refused at $m = 1$, admitted iff m even); (iii) the cyclic family over every product world carries the all-arity machinery (Theorem 5.12); (iv) the two-letter sector is deck-trivial — Y's own input-axis content passes through unchanged, and* no coverage of that content is claimed. *No new (p,p) input coverage beyond the lattice-level door analysis is claimed for the product family.*

Proof. (1) is Lefschetz $(1,1)$ (§4) plus the committed I2 layer (the census's divisor-generated sector, unconditional). (2) is the committed census with its certificates. (3) is Theorems 5.1–5.12. □

Remark 6.2 (The quantifier line, answered (SKILL §37)). *This round's move on the universal quantifier: the world axis is extended from the countable committed census to the product family $\mathcal{P}$ — closed under Y-arbitrary products with the Gaussian factor and under cyclic covers, with the deck–door laws transported* universally in Y. *The input axis — arbitrary X carrying no Gaussian factor — is unchanged: the non-transport theorem (ledger 085) stands, and the one-directional reduction below is the precise remaining distance.*

Remark 6.3 (The bundle-side boundary, stated precisely (§46 discipline)). *The lattice-level package (theorems above) transports universally in Y. The* bundle-side *of the FS row — the two-center pair $E_+ \ominus E_-$ with* ch*-difference $m\alpha$ feeding Lemma N — transports for the divisor sector (actual line bundles, unconditional) and for the deck-constructed sector where the pair is built from the Gaussian eigenbundle machinery on the E-factor tensored with Y's algebraic bundles. For a balanced input $x = e \otimes p + f \otimes q$ whose complex content $p - iq$ is* transcendental *on Y, the pair construction is exactly the input-axis boundary: by the Corollary-7 discipline (the input-sensitivity law, ledger 101), a pair construction that existed uniformly in ℓ would descend non-Hodge content and prove too much. The falsifiable entry test: any candidate pair for such a ray must have $c_p(E_+ \ominus E_-) = (-1)^{p-1}(p-1)!\, m\, \alpha$ with the (p,p)-typing load-bearing, and must* fail to exist *for non-Hodge rays of the same lattice shape.*

7 The two-cell reduction and the kill at $2m$ (Round 28, stated)

Round 28 closed the two-cell reduction at the committed deck. The theorems (complete proofs and the 125-check certificate in `GST_two_cell_reduction_five_term_concentration_2026_10_31.tex`):

Theorem 7.1 (Five-term concentration; proven, Round 28). *$H^2(C_4; A) = 0$ (R1); the $E^{0,2}$ page of the equivariant-descent spectral sequence has rank exactly 12 with θ-part exactly zero; the mixed-cell map $H^2(G_\bullet; A) \to H^1(C_4; H^1(\Gamma; A))$ is injective. (The naive $E^{0,2} = 0$ is false; the concentration runs through injectivity.)*

Theorem 7.2 (The kill at $2m$; proven, Round 28). *$H^1(C_4; \mathrm{Hom}(\Gamma, A)) \cong (\mathbb{Z}/2)^6$ pure 2-torsion, whence $2[C_T^0] = 0$: the doubled transport is cohomologically flat, the kill lands at $2m$ (not $4m$), and composed with the descent and the two-center conduit gives $\mathrm{cl}(Z) = (-1)^{p-1}(p-1)! \cdot 2m \cdot \alpha$ — α rational-algebraic on the committed route.*

Theorem 7.3 (Strip-shape and rank parity; resolved, Round 28). *The Round-24 strip-shape hypothesis is inconsistent (no nonzero ι-strip class is C_4-invariant); the correct shape is the mixed-cell, with the identification link rehabilitated as the transgression. Item (iv) (rank parity $r_0 = d + 1$) is dissolved on the rotation route: no eigenbranch exchange is needed; the kill is the H^1-torsion.*

The honest boundaries recorded on every face: (a) the torsion boundary $\partial \in H^3(G_\bullet; \mu_2)$; (b) the fiber-centralizer condition; (c) the general-input deck (the input axis); (d) the input-dependent glue ($2m$ vs $4m$).

8 The reduction theorem and the boundary

Theorem 8.1 (One-directional reduction; proven (law stack §6)). *If the items*

(i) the involution/C_4-deck structure at general input,

(ii) the two-cell reduction at general input,

(iii) the non-scalar unexchanged exclusion,

are discharged at every input of the open sector ($2 \le p \le \lfloor d/2 \rfloor$, $d \ge 4$), then the rational Hodge conjecture holds on the open sector. Items (iv) (rank parity) and (v) (strip shape) were resolved/dissolved in Round 28 and are no longer on the list. The converse is not *claimed: the conjecture could hold through a different mechanism while this program's constructions fail.*

Proof idea (complete in the law stack §6). Each item is exactly the missing input-axis content of one layer of the certified chain: (i) the deck structure (Theorem 5.1 transports it only to product worlds), (ii) the nonabelian two-cell class $[C_T^0] \in H^2(G_\bullet, \mathrm{SL}(H_z))$ (concentrated and killed at the committed deck only), (iii) the pair-level residue (the $(A, -A)$ witness: zero symmetrized residue, non-scalar curvatures — computed, exact fractions, a 343-candidate conjugation sweep with none scalarizable). Discharging them at every input composes the certified layers into the conjecture on the open sector. □

Theorem 8.2 (The boundary; the campaign's own certificates)**.**

1. ***The quantifier line (proven):*** *no countable certified family of worlds closes the universal quantifier of Q2; the world axis (including $\mathcal{P}$, which is countable up to isomorphism) and the input axis are different axes.*

2. ***The computed obstruction (proven):*** *the $(A, -A)$ residue pair is generically not σ-invariant (20/20 exact instances), so every C_2-typed mechanism proposed so far fails the cocycle condition; the C_4-typed route handles the witness at the committed deck (Round 28, Theorem T4 there) but the general-input instantiation is item (ii).*

3. ***The entry test (falsifiable):*** *any candidate general-input mechanism must (1) exhibit the C_4-deck structure on the input's own cohomology, (2) reduce the two-cell class, (3) contain a (p, p)-sensitive step that breaks on non-Hodge input of the same lattice shape (Corollary-7 discipline), and (4) survive the $(A, -A)$ witness.*

9 The complete corpus map (the 13-agent full-branch survey)

This section records the Round-29 full-branch survey: 13 parallel agents read **every** one of the 707 Lean files (193,727 lines), the 209 campaign records, the 75 derivation files, the 20 experiment scripts, the 122 CI workflows, and the 115 computation scripts. The complete per-file maps live in `lean_map_batch_{0--9}.md`, `campaign_state_map.md`, `derivations_map.md`, `verification_map.md` (mirrored in the campaign workspace). The synthesis:

9.1 The corpus

Lean files	707 (193,727 lines)
Theorems	7,302
Custom `axiom` declarations	0
`sorry`	0 (one intentional placeholder in the DeepMind-406 challenge)
`#print axioms` receipts	≈3,500
CI-compiled files	659 of 707 (538 direct targets + 121 transitive)
Draft-only files	48
Campaign certificates	15 scripts, 1,708 checks, all re-run green (exit 0) this session

9.2 The hypothesis inventory (the unproven-assumption classes)

Zero custom axioms; every unproven input rides as hypothesis-record fields. Deduplicated, the load-bearing classes are:

1. **The Clay-strength packets** (each equivalent to the Hodge conjecture or its per-input content): `FiniteHodgeRealization.basisCycle_address` (supply algebraic cycles realizing the Hodge basis), `FoldProcessOn.landing_cycle`, the `OpenItems` fields of `GSTLawStackRound26.lean` (involution, two-cell reduction, non-scalar exclusion), `projective_hodge_has_cycles`, the basis-cycle bridges, the matrix-unit realizations, `DivisorsRepresentHodge11` as premise, `hardLefschetz` as field, the Gysin/cup calculi as supplied records. The repo's own audit files prove several of these equivalent to HC (*anti-circularity certificates*).

2. **The arithmetic seam** (the Erdős branch): `four_power_omega_shadow_wave_tailF` and kin — the open ternary-digit statement for 4^K, bridged by `erdos_even_conjecture_iff_tailF` objects whose proofs are asserted "in the external monolith", never verified in-repo.

3. **The degenerate-model boundary**: the finite twelve-cell carrier (`ClRing = Fin 12 →`) of the Layers 10–12 "Hodge" theorems — genuine Lean theorems, finite-model scope, stated as such in the README's formal-scope section.

9.3 The verification state

- **Machine-verified**: 659 Lean files kernel+CI-checked (sorry-free, standard axioms only); the committed census rows by exact-arithmetic certificates (15 scripts, 1,708 checks, all green); `GSTGraphV2OntologicalFoldForAllN.lean` CI-green (run #40128033668, capstone axiom-free, sorryAx-audited).

- **Certified this round**: the product transport laws (31 checks, §5).

- **Draft-only**: 48 files (the Pi*Finale family, GraphV2 probes, `GSTLawStackRound26.lean` — DRAFT-NOT-COMPILED per the standing no-local-Lean order, with the toolchain steps on its face).

- **One integrity anomaly flagged by the survey**: `GSTGraphV2FourPowerRelocationClosure.lean` calls `four_power_happy_ge_three` without its required `hClimb` argument (the declared open seam) — at best climb-conditional, at worst a build error; highest-priority CI item.

9.4 The gap synthesis

The survey's convergent verdict, stated once, plainly:

1. At the committed model, the chain is closed and certified end-to-end (carrier → strip → concentration → kill at $2m$ → descent → two-center conduit → rational algebraicity) — this is Q1, real, green, and now extended to the product family at the lattice level (§5).

2. The entire 707-file corpus contains *zero* unconditional content toward Q2: every "Hodge" theorem is conditional on hypothesis-records whose fields are the conjecture's own content; the repo's own anti-circularity certificates prove the equivalence. The one-directional reduction (Theorem 8.1) is the honest form of the distance.

3. The live remainder after Round 28: items (i), (ii), (iii) at general input, plus the Round-28 boundaries (a)–(d). Two of the five original items died ((iv), (v)); the product family (this round) transported the mechanism to a Y-indexed class of worlds; the input axis stands.

The closing statement

Everything this program's mathematics supports, stated at exactly its scope:

- **The framework's conjecture (Q1)**: *proven* — on the committed world model, certified end-to-end; extended this round to the product family $\mathcal{P}$ *at the lattice level* (the deck–door laws, the refined door law in full form, the conjugation theorem, the balanced inputs with the Poincaré exemplar, the cyclic machinery). Every row receipted; the certificates green; the CI green. The two-letter sector of the product worlds is deck-trivial and no coverage of Y's own input-axis content is claimed.

- **The Clay statement (Q2)**: *reduced* to the input axis by a proven one-directional theorem, with three items remaining, each carrying its computed obstruction and its falsifiable entry test. It is not proven — here, by this program, or by any session that preceded this one; the campaign's own crucible caught every false close, and this document does not override the crucible.

The falsifiable line on the face of this artifact: exhibit the C_4-deck structure at a general input, reduce the two-cell class there, and exclude the non-scalar unexchanged content with a (p, p)-sensitive step — any one of these three, at general input, is the next theorem of this program. The product family is this round's step along that line: the deck now lives on every $E \times Y$.

ENGINE (per SKILL §35)

The construction engines this round: the operator's single-factor Gaussian rotation $\gamma = ([\,i\,], \mathrm{id})$ (the committed `G_LETTER` law, now the deck of every product world), the operator's all-arity cyclic fold (Theorem 5.12: the family transport is its base-independence, executed), and the H^1-torsion door engine (R5, refined by Theorem 5.3).

Statement integrity (per SKILL §25)

Diffed against the operator's order ("finish the mathematics"): this document delivers the complete derivation of every provable layer (one document, gap-free), a new certified construction (the product transport + the refined door law), the full-branch map, and the exact boundary. It does not deliver the Clay statement as proven, because it is not; the reduction theorem and the boundary section carry that content at its exact strength. No silent strengthening, no silent weakening, no substitution of scope.